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Title

TESI DI LAUREA MAGISTRALE IN
COMPUTER SCIENCE AND ENGINEERING - INGEGNERIA
XXXXXXX

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Abstract

Here goes the Abstract in English of your thesis followed by a list of keywords. The Abstract is a concise summary of the content of the thesis (single page of text) and a guide to the most important contributions included in your thesis. The Abstract is the very last thing you write. It should be a self-contained text and should be clear to someone who hasn't (yet) read the whole manuscript. The Abstract should contain the answers to the main scientific questions that have been addressed in your thesis. It needs to summarize the adopted motivations and the adopted methodological approach as well as the findings of your work and their relevance and impact. The Abstract is the part appearing in the record of your thesis inside POLITesi, the Digital Archive of PhD and Master Theses (Laurea Magistrale) of Politecnico di Milano. The Abstract will be followed by a list of four to six keywords. Keywords are a tool to help indexers and search engines to find relevant documents. To be relevant and effective, keywords must be chosen carefully. They should represent the content of your work and be specific to your field or sub-field. Keywords may be a single word or two to four words.

Keywords: here, the keywords, of your thesis

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Introduction

This document is intended to be both an example of the Polimi L^AT_EX template for Master Theses, as well as a short introduction to its use. It is not intended to be a general introduction to L^AT_EX itself, and the reader is assumed to be familiar with the basics of creating and compiling L^AT_EX documents (see [4, 6]).

The cover page of the thesis must contain all the relevant information: title of the thesis, name of the Study Programme and School, name of the author, student ID number, name of the supervisor, name(s) of the co-supervisor(s) (if any), academic year. The above information are provided by filling all the entries in the command `\puttitle{}` in the title page section of this template.

Be sure to select a title that is meaningful. It should contain important keywords to be identified by indexer. Keep the title as concise as possible and comprehensible even to people who are not experts in your field. The title has to be chosen at the end of your work so that it accurately captures the main subject of the manuscript.

Since a thesis might be a substantial document, it is convenient to break it into chapters. You can create a new chapter as done in this template by simply using the following command

```
\chapter{Title of the chapter}
```

followed by the body text.

Especially for long manuscripts, it is recommended to give each chapter its own file. In this case, you write your chapter in a separated `chapter_n.tex` file and then include it in the main file with the following command

```
\input{chapter_n.tex}
```

It is recommended to give a label to each chapter by using the command

```
\label{ch:chapter_name}%
```

where the argument is just a text string that you'll use to reference that part as follows:

Chapter ?? contains AN INTRODUCTION TO

If necessary, an unnumbered chapter can be created by

`\chapter*{Title of the unnumbered chapter}`

1 | State of the art

In this chapter additional useful information are reported.

1.1. Sections and subsections

Chapters are typically subdivided into sections and subsections, and, optionally, sub-subsections, paragraphs and subparagraphs. All can have a title, but only sections and subsections are numbered. A new section is created by the command

```
\section{Title of the section}
```

The numbering can be turned off by using `\section*{}`.

A new subsection is created by the command

```
\subsection{Title of the subsection}
```

and, similarly, the numbering can be turned off by adding an asterisk as follows

```
\subsection*{}
```

1.2. Equations

This section gives some examples of writing mathematical equations in your thesis.

Maxwell's equations read:

$$\left\{ \begin{array}{l} \nabla \cdot \mathbf{D} = \rho, \\ \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = \mathbf{0}, \\ \nabla \cdot \mathbf{B} = 0, \\ \nabla \times \mathbf{H} - \frac{\partial \mathbf{D}}{\partial t} = \mathbf{J}. \end{array} \right. \quad \begin{array}{l} (1.1a) \\ (1.1b) \\ (1.1c) \\ (1.1d) \end{array}$$

Equation (1.1) is automatically labeled by `cleveref`, as well as Equation (1.1a) and Equation (1.1c). Thanks to the `cleveref` package, there is no need to use `\eqref`.

Remember that Equations have to be numbered only if they are referenced in the text.

Equations (1.2), (1.3), (1.4), and (1.5) show again Maxwell's equations without brace:

$$\nabla \cdot \mathbf{D} = \rho, \quad (1.2)$$

$$\nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = \mathbf{0}, \quad (1.3)$$

$$\nabla \cdot \mathbf{B} = 0, \quad (1.4)$$

$$\nabla \times \mathbf{H} - \frac{\partial \mathbf{D}}{\partial t} = \mathbf{J}. \quad (1.5)$$

Equation (1.6) is the same as before, but with just one label:

$$\left\{ \begin{array}{l} \nabla \cdot \mathbf{D} = \rho, \\ \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = \mathbf{0}, \\ \nabla \cdot \mathbf{B} = 0, \\ \nabla \times \mathbf{H} - \frac{\partial \mathbf{D}}{\partial t} = \mathbf{J}. \end{array} \right. \quad (1.6)$$

1.3. Figures, Tables and Algorithms

Figures, Tables and Algorithms have to contain a Caption that describe their content, and have to be properly referred in the text.

1.3.1. Figures

For including pictures in your text you can use `TikZ` for high-quality hand-made figures, or just include them as usual with the command

```
\includegraphics[options]{filename.xxx}
```

Here xxx is the correct format, e.g. `.png`, `.jpg`, `.eps`,



Figure 1.1: Caption of the Figure to appear in the List of Figures.

Thanks to the `\subfloat` command, a single figure, such as Figure 1.1, can contain multiple sub-figures with their own caption and label, e.g. Figure 1.2a and Figure 1.2b.



Figure 1.2: This is a very long caption you don't want to appear in the List of Figures.

1.3.2. Tables

Within the environments `table` and `tabular` you can create very fancy tables as the one shown in Table 1.1.

Title of Table (optional)			
	column 1	column 2	column 3
row 1	1	2	3
row 2	α	β	γ
row 3	alpha	beta	gamma

Table 1.1: Caption of the Table to appear in the List of Tables.

You can also consider to highlight selected columns or rows in order to make tables more

readable. Moreover, with the use of `table*` and the option `bp` it is possible to align them at the bottom of the page. One example is presented in Table 1.2.

	column1	column2	column3	column4	column5	column6
row1	1	2	3	4	5	6
row2	a	b	c	d	e	f
row3	α	β	γ	δ	ϕ	ω
row4	alpha	beta	gamma	delta	phi	omega

Table 1.2: Highlighting the columns

	column1	column2	column3	column4	column5	column6
row1	1	2	3	4	5	6
row2	a	b	c	d	e	f
row3	α	β	γ	δ	ϕ	ω
row4	alpha	beta	gamma	delta	phi	omega

Table 1.3: Highlighting the rows

1.3.3. Algorithms

Pseudo-algorithms can be written in $\text{\LaTeX}$ with the `algorithm` and `algorithmic` packages. An example is shown in Algorithm 1.1.

Algorithm 1.1	Name of the Algorithm
1:	Initial instructions
2:	for <i>for – condition</i> do
3:	Some instructions
4:	if <i>if – condition</i> then
5:	Some other instructions
6:	end if
7:	end for
8:	while <i>while – condition</i> do
9:	Some further instructions
10:	end while
11:	Final instructions

1.4. Theorems, propositions and lists

1.4.1. Theorems

Theorems have to be formatted as:

Theorem 1.1. *Write here your theorem.*

Proof. If useful you can report here the proof.

1.4.2. Propositions

Propositions have to be formatted as:

Proposition 1.1. *Write here your proposition.*

1.4.3. Lists

How to insert itemized lists:

- first item;
- second item.

How to insert numbered lists:

1. first item;
2. second item.

1.5. Use of copyrighted material

Each student is responsible for obtaining copyright permissions, if necessary, to include published material in the thesis. This applies typically to third-party material published by someone else.

1.6. Plagiarism

You have to be sure to respect the rules on Copyright and avoid an involuntary plagiarism. It is allowed to take other persons' ideas only if the author and his original work are clearly mentioned. As stated in the Code of Ethics and Conduct, Politecnico di Milano *promotes the integrity of research, condemns manipulation and the infringement of*

intellectual property, and gives opportunity to all those who carry out research activities to have an adequate training on ethical conduct and integrity while doing research. To be sure to respect the copyright rules, read the guides on Copyright legislation and citation styles available at:

<https://www.biblio.polimi.it/en/tools/courses-and-tutorials>

You can also attend the courses which are periodically organized on "Bibliographic citations and bibliography management".

1.7. Bibliography and citations

Your thesis must contain a suitable Bibliography which lists all the sources consulted on developing the work. The list of references is placed at the end of the manuscript after the chapter containing the conclusions. We suggest to use the BibTeX package and save the bibliographic references in the file `Thesis_bibliography.bib`. This is indeed a database containing all the information about the references. To cite in your manuscript, use the `\cite{}` command as follows:

Here is how you cite bibliography entries: [2], or multiple ones at once: [3, 5].

The bibliography and list of references are generated automatically by running BibTeX [1].

2 | Proposed solution

2.1. Problem formulation

Consider a simulated TR-SPAD sensor observing a MNIST digit over an acquisition interval $[0, T_{\text{acq}}]$. The resulting photon-event stream is represented as

$$\mathcal{S} = (e_i)_{i=1}^N, \quad e_i = (x_i, y_i, t_i),$$

Where N is the number of detected photon-events, $(x_i, y_i) \in \{0, \dots, 27\}^2$ are the pixel coordinates of the event, and t_i is the arrival time measured in seconds

Each recording has a ground-truth digit label $v \in \{0, \dots, 9\}$.

The objective is to develop a CNN-based pipeline that correctly classifies MNIST digits from TR-SPAD event streams and generalizes to previously unseen recordings.

In particular we aim to obtain a correct classification while avoiding separate conventional image-reconstruction methods prior the usage of the CNN. We try to achieve this by directly using temporal information contained in the event stream.

2.2. Dataloading and photon binning

The event stream is first reorganized to support parallel processing on the GPU. Each recording is represented by a dense tensor of dimensions $[H, W, T, K]$, where H and W are the sensor dimensions. The acquisition interval T_{acq} is divided into T equal temporal bins, each of duration

$$\Delta t_{\text{bin}} = \frac{T_{\text{acq}}}{T}.$$

Each photon event is assigned to a bin according to its timestamp and pixel coordinates. The stored value is its interarrival time, i.e. the time passed since the previous photon detection, for the same pixel, across bin boundaries. Multiple events at the same pixel within a bin occupy successive slots along the K dimension.

The number of slots K represents the maximum number of events occurring at any pixel

within a single bin. Unused slots are padded with zeros, and a binary mask distinguishes valid events from padding.

The resulting tensor and mask are passed to the sensor-proximal layer together with the time window represented by each bin. This differentiates the proposed solution from QIS methods by preserving individual photon timing and allows parallel processing across pixels.

2.3. Sensor-proximal Layer

Photon arrival model

TR-SPAD sensors provide the pixel coordinates and arrival time of each detected photon with microsecond precision. At pixel location $\mathbf{p} = (x, y)$, let $t_i(\mathbf{p})$ denote the arrival time of the i -th photon. The corresponding interarrival time is

$$\Delta t_i(\mathbf{p}) = t_i(\mathbf{p}) - t_{i-1}(\mathbf{p}), \quad (2.1)$$

where the acquisition start is used as the reference for the first detection.

Considering a constant photon arrival rate $\lambda(\mathbf{p})$, interarrival times can be modeled as conditionally independent exponential random variables, with probability density

$$f(\Delta t_i(\mathbf{p}) \mid \lambda(\mathbf{p})) = \lambda(\mathbf{p}) e^{-\lambda(\mathbf{p}) \Delta t_i(\mathbf{p})}. \quad (2.2)$$

BOCPD

For each pixel location, up to eight run-length hypotheses, or forecasters, are maintained in memory. Each forecaster represents a possible time of the most recent change in photon arrival rate. The uncertainty about these rates is represented by a Gamma distribution with shape parameter α and rate parameter β :

$$f(\lambda \mid \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{(-\beta\lambda)}, \quad \lambda > 0. \quad (2.3)$$

The Gamma distribution is conjugate to the exponential likelihood, so the posterior remains Gamma.

The likelihood is obtained by using the observed photon interval to integrate over the

possible rates

$$L_s(\Delta t_i) = \int_0^\infty \lambda e^{-\lambda \Delta t_i} f(\lambda \mid \alpha_s, \beta_s) d\lambda = \frac{\alpha_s \beta_s^{\alpha_s}}{(\beta_s + \Delta t_i)^{\alpha_s+1}}. \quad (2.4)$$

For a continuing hypothesis, each observed interarrival time updates its parameters as $\alpha \leftarrow \alpha + 1$ and $\beta \leftarrow \beta + \Delta t_i$.

The forecasters weights ν are then updated as

$$\nu_s = (1 - \gamma) \nu_s L \quad (2.5)$$

where $0 < \gamma < 1$ controls the responsivity of run length modeling.

Adaptive exponential integration

This is center of this approach, when a net is processing a video, it is essential to judge when it is proper to accept new photon information faster, an object moving rapidly for example, to limit blurring, and when to integrate longer, decreasing noise as a result. Thanks to the proposed method, there is no need to decide beforehand on an integration parameter for the whole video sensor, it is tuned at runtime at the pixel level.

To this end we first calculate for each pixel the geometric average of the run lengths given by the forecasters

$$\hat{r}_n(\mathbf{p}) = \exp \left(\sum_{s=1}^S \nu_{s,n}(\mathbf{p}) \ln(1 + r_{s,n}(\mathbf{p})) \right) - 1, \quad (2.6)$$

this returns an approximation of how long the current photon arrival rate has been constant. Compared with the weighted arithmetic mean, this expression reduces the influence of hypotheses with large run lengths.

We apply spatial minimum pooling on the obtained $\hat{r}_n$ over the local neighborhood so that the pixel with lowest run length influences the surrounding pixels.

However, $r_{s,n}$ represents the age of the forecaster in units of temporal bins, which means that the approximation is also in bin units. We need then to convert $\hat{r}_n$ into seconds with $\tau_n = \hat{r}_n \Delta t_{\text{bin}}$

The integration time is calculated at the beginning of each bin and held fixed throughout that bin. BOCPD updates performed during the bin influence the integration time of the

following bin.

The sensor-proximal layer then adaptively integrates the photon-events at each pixel by performing normalized exponential smoothing:

$$I_t = \omega_t I_{t-1} + \frac{1}{\tau_n} \quad (2.7)$$

where ω is calculated as

$$\omega_t = e^{-\delta t / \tau_n} \quad (2.8)$$

with δt being the time since the last update.

It is important to note that adaptive integration and BOCPD update happens only for the K slots that contain a photon-event. Decay still needs to be applied to all of the pixels even when photons are not hitting the sensors.

As a solution we can make use of the binning process described in section 2.2. Since each bin corresponds to an equal slice of acquisition time, they are perfect as they provide a common time reference for the decay. After integrating new photon events for the time bin, every pixel (even the ones hit by a photon) is decayed through the remaining interval since its last accumulator update.

2.4. Subsampling and CNN

A temporal subsampling factor D determines how frequently the integration values are recorded and used for classification. After every D raw bins, the current accumulator is saved as a spatial readout, corresponding to a readout interval of $D\Delta t_{\text{bin}}$. The BOCPD and adaptive integrator continue processing all photon events between readouts, and their states are retained. Subsampling therefore reduces the number of readouts processed by the rest of the network.

Each normalized readout is treated as a single-channel 28×28 image and passed through two convolutional layers. Both layers use 3×3 kernels with stride one and no padding, producing 64 and 128 feature channels, respectively. Each convolution is followed by a ReLU activation. The spatial dimensions decrease from 28×28 to 26×26 and then to 24×24 .

The resulting feature maps are passed in chronological order to the SSD state-space layer, together with their readout timestamps. This layer maintains a temporal state that combines the current features with information from previous readouts, taking into account

the elapsed time between them. It therefore provides temporal processing of the features extracted by the CNN.

After the SSD layer, global average pooling reduces each feature map to a single value per channel, producing a 128-dimensional vector for each readout. During training, these vectors are averaged across the selected readouts of the recording. A final linear layer maps the resulting vector to ten digit-class scores.

During evaluation, readouts are processed successively while retaining the SSD state within each digit recording. At each readout, the pooled feature vectors available in the configured history window are averaged to produce an updated prediction. The temporal state and feature history are reset between independent digit recordings.

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A | Appendix A

If you need to include an appendix to support the research in your thesis, you can place it at the end of the manuscript. An appendix contains supplementary material (figures, tables, data, codes, mathematical proofs, surveys, . . .) which supplement the main results contained in the previous chapters.

B | Appendix B

It may be necessary to include another appendix to better organize the presentation of supplementary material.

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List of Symbols

Variable	Description	SI unit
$\boldsymbol{u}$	solid displacement	m
$\boldsymbol{u}_f$	fluid displacement	m

Acknowledgements

Here you might want to acknowledge someone.

